

Operators

RTL is mostly expressions: AND gates, compares, shifts, and bus packing

Bitwise, logical, and reductions

- Bitwise operators, ampersand, pipe, caret, tilde, work per bit on vectors of any width
- Logical operators
- All bits ANDed, any bit ORed, or parity XOR
- Concatenation braces glue vectors left to right; replication repeats a pattern
- Shifts move bits with zero fill or, for arithmetic right shift, sign-bit fill

Browser lab

igital Design and Verification Platform

HomeToolsSV operators

Challenge created

Quiz: & vs &&: For multi-bit vectors, & is ____ while && is ____.

☐ bitwise / logical (1-bit)

☐ logical / bitwise

☐ the same

☐ only for X/Z

Show hint

Check

Next

Load challenge setup

Idle

Quiz: & vs &&

Quiz: && width

Quiz: ~ vs !

Quiz: reduction &

Quiz: reduction |

Quiz: concat

Quiz: replicate

Quiz: <<

Quiz: >>

Quiz: ==

Run: bitwise &

Run: logical &&

Run: && with zero

Run: bitwise ~

Run: logical !

Run: & reduction

Run: concat

Run: replicate

Run: <<

Run: >>

See & vs &&

Quiz: parity ^

Operands & operators

Load starter example

A

B

N (SHIFT / REPLICATE)

4'b1010

4'b1100

1

&

|

^

~

&&

||

!

&()

|()

^()

{,}

{n{}}

<<

>>

>>>

==

===

4'b1010 & 4'b1100

1000 (4'b)

Bitwise AND – per bit – per-bit AND

A=1010 · B=1100

Bitwise &

1000

Logical &&

1

Literals via HDL parseLiteral

SV operators lab starter

learn_verilog — sv operators

Real Verilog practice

```
wsl - module05-sv-operators/examples/navigation

# Track A - real Unix

# real Verilog session (Track A)

$ cat examples/operators/operators.v
// operators.v - bitwise vs logical, concat, reduction
module operator_demo(
    input wire [3:0] a,
    input wire [3:0] b,
    output wire [3:0] bw_and,
    output wire      log_and,
    output wire [7:0] cat_ab,
    output wire      all_ones
);
    assign bw_and   = a & b;
    assign log_and  = a && b;
    assign cat_ab   = {a, b};
    assign all_ones = &a;
endmodule

$ iverilog -t null examples/operators/operators.v
(syntax check passed)
```

Real Verilog — operator demo

Pitfalls to watch

- Do not swap bitwise and logical AND
- Do not confuse concat order: left operand becomes the upper bits
- Arithmetic right shift fills with the sign bit; logical right shift fills with zero
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load the starter and finish a few challenges
- On real Verilog
- When you are ready